Department of Information Science and Engineering

Cross-Domain Rumor Verification: Detecting Informed Trading Footprints via Sequential Decision-Making

Team Number: 31

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AGENDA

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- Literature Gap
- Motivation
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INTRODUCTION

- Rumors appear on Reddit hours before official news — a few real, most noise.
- The stock often moves before the announcement becomes public.
- Verification is a timing problem: waiting adds evidence but spends the advantage.
- Our system re-decides every hour: WAIT, or COMMIT to TRUE or FALSE.
- Evidence spans two domains — Reddit activity and hourly price and volume.

LITERATURE GAP

- Existing rumor detection (Castillo 2011, Ma 2016, Zubiaga 2018) classifies a post once, at a fixed cut-off.
- It never chooses when to decide, and cannot abstain when evidence is thin.
- Social signals only — the market's reaction to the claim goes unused.
- Financial NLP predicts returns after news, not veracity before it.
- Earliness is a by-product of the chosen horizon, never an optimised objective.
- No dataset links a rumor's first post to the document that later confirms it.

MOTIVATION

- A verdict that arrives after the press release has no value.
- Abnormal price and volume before an announcement is a footprint of informed trading.
- The real trade-off is accuracy against hours saved.
- A wrong early verdict costs more than a slow one, so the agent may keep waiting.
- Free-tier data keeps the whole study reproducible on student hardware.
- Purpose is market-integrity monitoring — not trading advice.

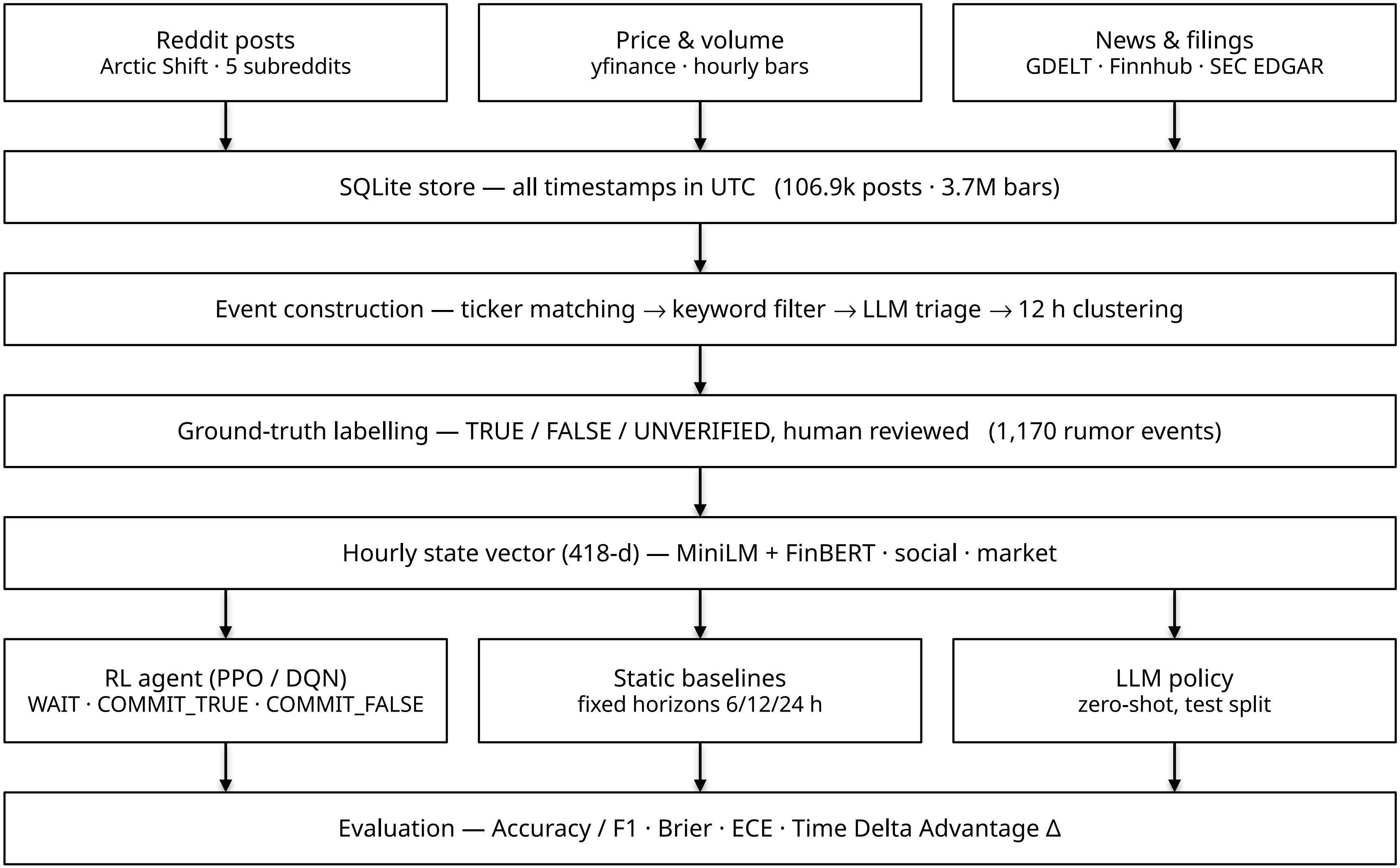
PROBLEM STATEMENT

- Input: a rumor event — ticker, claim, first-post time t_0 — with evidence arriving hourly.
- Each hour the agent picks one action: WAIT, COMMIT_TRUE or COMMIT_FALSE.
- Ground truth: t_official — the first SEC filing or whitelisted article within 72 h.
- Goal: be correct, and be early — maximise Δ = t_official − t_commit.
- Constraint: features at step t use only data timestamped $\leq t_0 + t$.
- Scope: binary veracity, US-listed tickers, 48-hour decision horizon.

OBJECTIVES

1. Build a labelled dataset of rumor events with confirmation timestamps.
2. Turn each event into an hourly state: text, social and market features.
3. Model verification as a three-action MDP; train with PPO and DQN.
4. Compare against fixed-horizon baselines and a zero-shot LLM policy.
5. Evaluate accuracy/F1, calibration (Brier, ECE) and Time Delta Advantage.
6. Demonstrate hour-by-hour replay of an event in a dashboard.

METHODOLOGY



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THANK YOU